Exploring root lengths

Describe center and spread using invented root lengths of separate seedlings measured once in mm. These practice data teach spreadsheet skills and provide no evidence about real plants.

Part 1: Explore the observations

- **Worksheet:** Roots

Step 1: Inspect the supplied data

- **Read:** Each number is one invented root length in millimeters.
- **Select:** **`A2:A6`**
- **Compare:** Find the shortest and longest root lengths. Both describe individual roots.
- **Explain:** Tell a classmate what one row represents and what the unit means.
- **Read:** Summary headings are in **`D1:E1`**; the three Value cells start empty.

Part 2: Summarize root lengths

- **Worksheet:** Roots

Step 1: Count the observations

- **Read:** Sample size is the number of numerical observations, not their sum.
- **Calculate:** In cell E2, enter `=COUNT(A2:A6)`.
- **Check:** Compare the result with the number of supplied root lengths.

Step 2: Calculate the mean

- **Read:** The arithmetic mean adds the root lengths and divides by the sample size.
- **Calculate:** In cell E3, enter `=AVERAGE(A2:A6)`.
- **Explain:** The mean describes the center of these observations in millimeters.

Step 3: Describe the spread

- **Read:** Sample standard deviation describes spread around the mean in millimeters. Use `STDEV.S`; its calculation uses the number of observations minus one.
- **Calculate:** In cell E4, enter `=STDEV.S(A2:A6)`.
- **Check:** Display three decimal places; keep full stored precision.
- **Explain:** Describe mean and SD together: center and spread among root lengths. SD is neither sample size nor the accuracy of an individual measurement.